

# Trace Boggs

<https://www.linkedin.com/in/trace-boggs-5405412a7/>

<https://tboggs05.github.io/>

Email : [trace.boggs05@gmail.com](mailto:trace.boggs05@gmail.com)

Mobile : 580-699-1689

## EDUCATION

---

### Cameron University

Lawton, OK

- *Bachelors of Science in Computer Science; 4.0 GPA with Presidents Honor Roll*  
*& Bachelors of Arts in Mathematics; 4.0 GPA with Presidents Honor Roll.*

August 2023 – May, 2026

## RESEARCH EXPERIENCE

---

### Math Capstone Research on Newton-Like Methods

Cameron University

*Undergraduate Researcher*

August 2025-Present

- **Paper:** Assisted in writing a research paper titled *Developments on the Local Convergence of Newton-Like Methods in Normed Linear Spaces with Applications to Motion Problems*. Used LaTeX to write up the paper, designed applied examples of the theory, and helped in constructing the proofs of the theory.
- **Poster Presentation(Feb 2026):** Designed a research poster to be presented in February of 2026 at Oklahoma Research Day.
- **NCUR Presentation(April 2026):** Created a PowerPoint presentation to be presented in April 2026 at NCUR.

### Neural Networks Independent Study

Cameron University

*Undergraduate Researcher*

January 2025 – December 2025

- **Research Poster Presentation:** Conducted a literature review on the evolution of artificial intelligence and designed an educational research poster titled *Artificial Intelligence: Past, Present, and Future*, presented at Oklahoma Research Day 2025.
- **Live-Inference Emotion Recognition Model:** Developed a CNN-based human emotion recognition system using transfer learning with MobileNetV2, fine-tuned on the FER-2013 dataset. Investigated performance degradation and prediction instability when transitioning from static image evaluation to real-time webcam inference.

## WORK EXPERIENCE

---

### Brunswick Navico Group

Tulsa, Oklahoma

*Software Engineer Intern*

May 2025 - August 2025

- **Python GUI Development:** Built a desktop application enabling engineers to communicate with embedded devices over BLE.
- **Automated Integration Tests:** Implemented tests to validate device configuration, verify firmware responses, and catch incorrect setting updates.
- **Object-Oriented Architecture:** Designed modular classes for BLE communication, packet construction, and test execution to ensure maintainability.
- **UX & Logging:** Added progress bars, real-time updates, and structured log files to support QA and debugging.
- **Outcome:** Brought down regression testing time during development by over 95%, significantly increasing engineer productivity.

### Cameron University

Lawton, Oklahoma

*Mathematics Tutor*

January 2024 - Present

- **Tutored** students in Math and Computer Science subjects, including College Algebra, Trigonometry, Calculus, Programming, and Discrete Math.
- **Led both one-on-one and group tutoring sessions**, helping students with coursework, problem-solving strategies, and exam preparation, resulting in improved homework and exam performance.
- **Communicated** regularly with students to identify gaps in understanding, build strong study habits, and reinforce key concepts.

## PROJECTS

---

- **Euler's Method Calculator** Sep 2024  
*C++, Crow, HTML/CSS, JavaScript, SSL, Nginx*
  - **Description:** Developed a full-stack web application to numerically solve ODEs with Euler's Method, hosted on a Raspberry Pi and accessible at [eulerscalculator.net](http://eulerscalculator.net).
  - **Backend:** Built a C++ backend using Crow, implementing a RESTful API for all computations and optimizing frontend-backend communication.
  - **Error Handling:** Implemented robust validation and domain-checking for mathematical expressions.
  - **Deployment:** Configured SSL and reverse proxying with Nginx for secure public hosting.
- **LightStop Mobile Game** May 2024  
*C#, Unity, APIs*
  - **Description:** Developed and released a mobile game inspired by Cyclone/Pop-the-Lock, published on both iOS and Android.
  - **Features:** Integrated Unity Ads, implemented a live leaderboard, and designed gameplay mechanics and visual effects.
  - **Deployment:** Managed platform builds, store listings, and release pipeline for both platforms.
  - **Website:** [sites.google.com/view/lightstop/home](https://sites.google.com/view/lightstop/home).
- **University PLUS Scholarship Website** Ongoing  
*Leadership, Full-Stack Web-Dev*
  - **Description:** Leading a team of university ACM members to design and develop a web application for the university. Coordinating development tasks, implementing features, and ensuring project progress toward completion next semester.

## ACHIEVEMENTS

---

- **Regional University Baccalaureate Scholarship Recipient:** Received the institution's top academic scholarship, which included a full tuition waiver and a significant stipend based on academic merit.

## TECHNICAL SKILLS

---

- **Proficient:** C++, C, Unity/C#, Git, HTML/CSS/JS/PHP, LaTeX
- **Comfortable:** Java, Python, Arduino, TensorFlow, Flutter, SQL, MATLAB